

NGUYEN GIA HUY

Golang Backend Engineer

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PROFESSIONAL SUMMARY

Golang backend engineer with nearly three years of software engineering experience, building performance-critical services and tooling for core banking. Built a Golang account migration system on AWS ECS, Lambda, and RDS that cut a migration run from **2.5 hours to 27 minutes**, and rewrote a Gauge-based simulation testing framework in Golang so it finishes in **just over 50 seconds instead of 24 minutes**. Builds gRPC and gRPC-Gateway backends on Kafka, MongoDB, and Redis that serve **1.5K requests/second** at **80–100 ms** with **99.95% uptime**, and brings data-pipeline, cloud, and microservices platform experience from KDDI's au Denki and VNPAY Cloud.

TECHNICAL SKILLS

Golang: Go, concurrency (goroutines, channels), gRPC, gRPC-Gateway, Protocol Buffers, REST/JSON APIs, test frameworks
Messaging: Kafka, RabbitMQ
Data Stores: PostgreSQL, Amazon RDS, MongoDB, Redis, OpenSearch, ClickHouse
Microservices: Kong Gateway, Keycloak (RBAC), Docker, Kubernetes
Cloud: AWS (ECS, Lambda, EKS, RDS, S3, ECR, ELB)
Observability: OpenTelemetry, Prometheus, Grafana, ELK
Also used: Python (Thought Machine Vault Core Smart Contracts), SQL

WORK EXPERIENCE

Golang Backend Engineer, Core Banking (Middle Software Engineer) 03/2026 – Present
Galaxy FinX (Vikki Digital Bank) | *core banking platform built on Thought Machine Vault Core*

Account Migration System (Golang)

- Designed and built the bank's **account migration system in Golang** for the core banking platform, bringing a migration run down from **2.5 hours to 27 minutes (5x faster)**.
- Deployed the migration processing on AWS: containerized Go workloads on **Amazon ECS** alongside **AWS Lambda** functions, with **Amazon RDS** as the relational store.

Testing Infrastructure (Golang)

- Replaced the legacy Gauge-based **simulation testing framework** with a new framework written **entirely in Go** for Thought Machine Vault Core.
- Reduced simulation test execution from **24 minutes to just over 50 seconds (20x faster)**, turning a 24-minute wait into feedback in under a minute.

Core Banking Integration and Delivery

- Connected the Vault Core ledger to the bank's surrounding backend services so core banking operations and transaction workflows run across systems.
- Shipped the platform to AWS: a new Vault Core version running on **EKS** behind **ELB**, with images in **ECR** and data in **RDS**.
- Automated its release flow (build, test, deploy) with a **CI/CD pipeline**.
- Developed and maintained the Lending and Overdraft products in Vault Core's Python Smart Contracts: interest accrual, penalty and late-fee calculation, repayment, and settlement.

Software Engineer – Backend & Data Services

04/2025 – 03/2026

Vietlink (KDDI Agile Development Center Group) | *au Denki – large-scale electric utility platform serving millions of users*

- Restructured batch services around **parallel execution**, pipeline redesign, and more efficient data access patterns, raising processing throughput by **150%**.
- Ran ETL workloads as **Lambda** functions and **ECS** tasks over large-scale utility data, reading from and writing to S3, Athena, and RDS for analytics and operations.
- Moved the platform's search and analytics layer from **Elasticsearch to OpenSearch**.
- Reworked how data is ingested and indexed into OpenSearch, improving stability, consistency of indexed data, and query performance.
- Kept the services deployable and observable: Terraform and AWS SAM for infrastructure, CloudWatch and DataDog for health, failure, and anomaly signals.

Software Engineer

04/2024 – 04/2025

VNPAY | *VNPAY Cloud – cloud platform for fintech and enterprise customers*

- Worked inside a **100+ service** microservices ecosystem fronted by **Kong Gateway**, integrating and operating its service-level and network-level access control.
- Secured tenant boundaries for **20+ customer organizations** with **Keycloak** identities and role-based permissions.
- Ran the platform's Kubernetes runtime: **50+ production clusters** on OpenStack, managed through Cluster API for provisioning, scaling, and zero-downtime upgrades.
- Replaced roughly **80%** of manual data center provisioning and configuration work with automation.

Blockchain Developer

06/2023 – 09/2023

EsolLabs | *Innovaz Market Factory – cross-chain NFT and token platform*

- Wrote **event-driven services** that listen for on-chain events, process them, and keep off-chain databases and APIs in sync.
- Integrated the services with **5+ blockchain networks**, covering EVM chains (Ethereum, BSC, Polygon) and Move-based chains (Aptos, Sui).
- Made transaction handling consistent across every chain, preserving data integrity off-chain.
- Also developed and reviewed **15+** Solidity and Move smart contracts for DeFi and NFT features.

KEY PROJECT

DattingQ | Software Engineer

Dating platform

- Built the backend API layer with **gRPC** and **gRPC-Gateway**, exposing REST/JSON endpoints generated from Protocol Buffer contracts.
- Integrated **Kafka**, **MongoDB**, and **Redis** into the backend as its messaging and data layers.
- Sustained up to **1.5K requests/second** at **80–100 ms** API latency.
- Implemented multi-cluster failover and OpenTelemetry-based observability with Prometheus, Grafana, ELK, and ClickHouse, achieving **99.95% uptime**.

MeetQ | Software Engineer

AI-powered online meeting platform

- Built a real-time meeting platform on **LiveKit**, integrating OpenAI models for live transcription, voice translation, and meeting summarization.
- Designed the real-time translation pipeline and optimized conversational context handling for translation accuracy and coherence.

EDUCATION & CERTIFICATIONS

University of Science, Ho Chi Minh City (HCMUS)

B.Sc. Information Technology – Knowledge Engineering

DevOps on AWS Specialization | Coursera